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Intelligent Sign Language Recognition for Real-Time Text Conversion to Aid Speech and Hearing Impaired

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Abstract

This research presents an innovative system for converting sign language into text using an Arduino UNO-IoT platform and a custom-designed glove equipped with finger flex sensors. The flex sensors detect the bending angles of the fingers, generating real-time data corresponding to different sign language gestures. These signals are processed by the Arduino to interpret gestures, which are then displayed as text on a 16x2 LCD screen. The system is designed to improve communication for the hearing and speech impaired, providing a low-cost, portable, and user-friendly solution for converting hand gestures into readable text. Technical findings show that the system achieves an average accuracy of 85% for simple gestures and 75% for complex gestures, with consistent performance across multiple users. Results were analyzed in terms of angular displacement, sensor resistance, and gesture type recognition. The system effectively bridges the communication gap between sign language users and the general public, offering enhanced accessibility through IoT integration.

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Keywords: Gesture; Recognition; Sensors; Sign; Accessibility

1. Introduction

The development of real-time, accurate, and user-friendly sign language-to-text conversion systems holds significant potential for bridging communication gaps between the deaf, hard of hearing, and the general population. These systems translate hand gestures into readable text, enhancing accessibility in daily interactions, educational

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settings, and professional environments. For millions of individuals worldwide, sign language is the primary mode of communication. However, communication barriers between signers and non-signers often lead to social isolation and limited access to essential services. By enabling seamless communication, sign language-to-text conversion technology offers a transformative solution that empowers the deaf community and fosters inclusivity. Recent advancements in gesture recognition technology have paved the way for more sophisticated systems, including those that use 3D motion capture. A 3D sign language recognition (SLR) system offers a more comprehensive approach by capturing hand movements in 3D space. This method has been particularly effective in overcoming challenges such as lighting variations, motion blur, and occlusions, which are common in 2D video-based recognition systems. In this work, a 3D hand template was designed to capture key features of Indian Sign Language, and the resulting 3D models formed a robust dataset. By utilizing 3D motion capture, the system offers full gesture recognition with improved accuracy and reliability, surpassing traditional 2D methods [1]. This approach highlights the potential for developing more complete and efficient sign language recognition systems that can be adapted for use in diverse linguistic contexts.

In addition to motion capture technology, machine learning techniques are playing an increasingly important role in sign language recognition. Deep learning models, such as Convolutional Neural Networks (CNNs), have demonstrated remarkable success in classifying sign language gestures [2]. For example, by using the pre-trained SSD MobileNet V2 architecture and applying transfer learning, researchers have developed systems capable of recognizing sign language with high accuracy. This real-time gesture recognition model is not only useful for daily communication but also serves as a valuable tool for those learning sign language [3]. The model's ability to classify gestures consistently, even in challenging environments with uncontrolled backgrounds and varying light conditions, marks a significant step forward in sign language recognition. The integration of these advanced technologies into sign language-to-text conversion systems holds immense promise for creating accessible, inclusive, and intelligent solutions. By addressing the current limitations of gesture recognition, these systems can facilitate better communication for the deaf and hard of hearing, ultimately contributing to more inclusive societies [4].

The paper is organized as follows: Section II reviews the related work in the field of sign language recognition. Section III presents the proposed approach for gesture recognition. Section IV describes the physical setup, detailing the hardware configuration. Section V discusses the results and discussion, analyzing the system's performance, and finally, Section VI concludes the work with future directions.

2. Related Work

The development of smart gloves for hand gesture recognition has significantly advanced, particularly for aiding communication among individuals with speech and hearing impairments. Bhaskaran et al. (2016) proposed a smart glove system that translates Indian Sign Language gestures into speech using flex sensors and an Inertial Measurement Unit (IMU), demonstrating the system's versatility for applications beyond speech conversion, such as gaming and robotics [5]. Adewale and Olamiti (2018) introduced a machine learning-based system that converts sign language images to text and speech, providing an effective communication tool for non-signers interacting with deaf individuals [6]. Bharathi et al. (2021) used convolutional neural networks (CNN) to recognize ASL gestures with an accuracy of 88%, thus enhancing communication for the speech-impaired [7].

Kunjumon and Megalingam (2019) developed a system capable of recognizing Indian Sign Language and converting it into speech and text in English and Malayalam, showcasing the adaptability of the system for multiple languages [8]. Kumar et al. (2019) extended this concept by capturing non-verbal behaviors, such as hand and head movements, to facilitate communication between deaf-mute individuals and non-signers [9]. Peguda et al. (2022) explored speech-to-sign language translation for multiple Indian regional languages using a combination of speech recognition and text translation techniques, thereby improving accessibility for the hearing impaired [10].

Tiku et al. (2020) developed an Android application that converts real-time ASL gestures into text and speech, providing a user-friendly interface for non-signers to communicate effectively with sign language users [11]. Dutta et al. (2015) captured double-handed Indian Sign Language gestures using MATLAB, which processed the sign language images and converted them into speech and text [12]. Punsara et al. (2020) implemented an IoT-based system, combining wearable gloves with mobile applications to translate sign language gestures into digital voice,

enhancing communication between mute and non-mute individuals [13]. Muralidharan et al. (2022) proposed a glove with flex sensors that detect hand gestures as binary bits, offering an efficient, cost-effective alternative to vision-based approaches for sign language recognition [14]. Das et al. (2016) developed a smart glove using an Intel Galileo IoT kit for real-time gesture recognition, aimed at bridging communication gaps for the deaf and mute communities [15]. Rizwan et al. (2019) introduced a wireless glove system for ASL recognition, translating hand gestures into corresponding alphabets and words with data sent to a smartphone via Bluetooth [16]. Kumar et al. (2024) presented an IoT-based glove system designed for Indian Sign Language, converting detected gestures into speech using an Arduino-powered text-to-speech mechanism [17]. Sadek et al. (2017) designed a smart glove for Arabic Sign Language, using minimal sensors to translate complex gestures into spoken or written language [18]. Finally, Shukor et al. (2015) developed a glove system for Malaysian Sign Language using tilt sensors and an accelerometer, achieving an average recognition accuracy of 89% for alphabets, numbers, and words [19].

3. Proposed Approach

The development of this sign language recognition system begins with the selection of essential components such as the Arduino microcontroller, flex sensors, and power supply. Flex sensors are used to detect the bending of fingers, which correspond to various hand gestures in sign language. Once the components are selected, the wiring and circuit setup are completed by connecting the flex sensors to the Arduino, allowing the system to capture the sensor data accurately. Next, Arduino code is developed to read the flex sensor data. The code continuously monitors the bending of the sensors and sends the corresponding data to the computer. Following this, a Python environment is set up, where the necessary libraries for data acquisition and processing are installed. Python is used to receive the sensor data transmitted from the Arduino and prepare it for further processing.

Once the data is acquired, a model is trained to recognize different sign language gestures. This involves collecting gesture data corresponding to common words and letters in the sign language alphabet. The gesture data is processed to train the system for real-time gesture recognition. After training, the system is tested with various sign language gestures to ensure accuracy and reliability. Testing involves different users to validate the robustness of the system across various hand shapes and gesture variations. Finally, a simple user interface (UI) is designed using Python frameworks. The UI displays the recognized gesture as text in real-time, allowing the user to easily view the conversion of sign language gestures into readable text. Fig. 1 outlines the key stages of the sign language recognition system. It starts with detecting hand gestures through sensors, followed by sign recognition and generating the corresponding text output. The system loops back to continuously process new gestures.

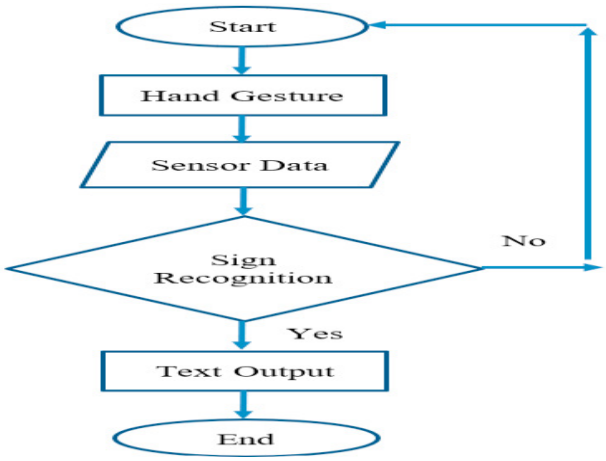

Fig. 1. Flowchart of sign language recognition system.

4. Proposed Approach

Fig. 2 shows the hardware setup used for the sign language to text conversion system. The setup includes an Arduino UNO microcontroller connected to a breadboard, a 16x2 LCD screen for displaying the output, and four flex sensors attached to the fingers of a glove. The flex sensors detect finger movements and send corresponding data to the microcontroller for processing and gesture recognition.

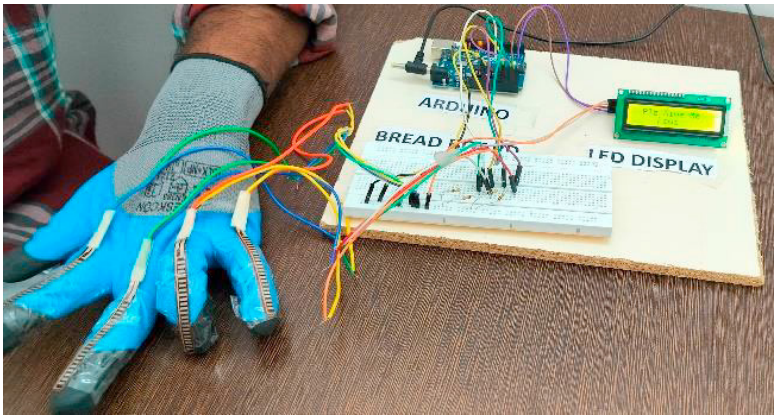

Fig. 2. Hardware setup of sign language to text conversion system.

In Fig. 3, the sign language gestures are translated into text and displayed on the 16x2 LCD screen. When a specific gesture is made using the glove's flex sensors, the corresponding message, such as "Give Me Water" or "Give Me Food," is displayed on the screen. This demonstrates how the system interprets hand gestures in real-time and converts them into readable text for effective communication.

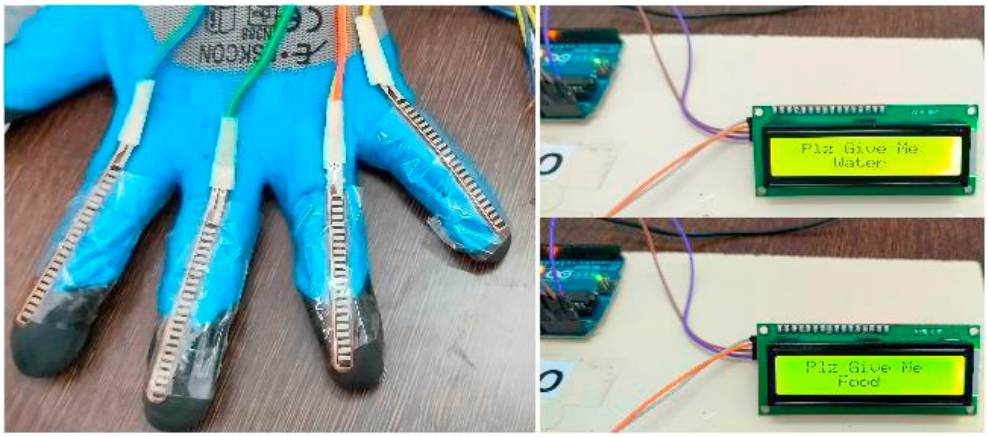

Fig. 3. Gesture recognition output on LCD screen.

The system begins by initializing the LCD display, serial communication, and flex sensors, displaying the message "Sign Language:" for two seconds before clearing the screen. In a continuous loop, the system reads the analog values from the flex sensors attached to the user's fingers. These values are converted into states based on predefined thresholds: "not bent," "partially bent," or "fully bent." Using the combination of sensor states, the corresponding sign language letter is identified from predefined mappings. The detected letter is then displayed in real-time on the LCD

screen while simultaneously printing the letter and sensor data to the serial monitor for debugging. A short delay ensures stability between gestures, and the system continues detecting and displaying gestures in real-time until it is manually stopped.

5. Results And Discussion

Fig. 4 illustrates the relationship between the bend angle and resistance for flex sensors. As the angular displacement increases from 0° to 90°, the resistance increases exponentially, from 10 KΩ to 80 KΩ. The flex sensors detect varying finger positions, which is crucial for capturing precise gestures in sign language. The data demonstrates how different angular displacements are translated into electrical resistance, allowing the system to interpret hand gestures.

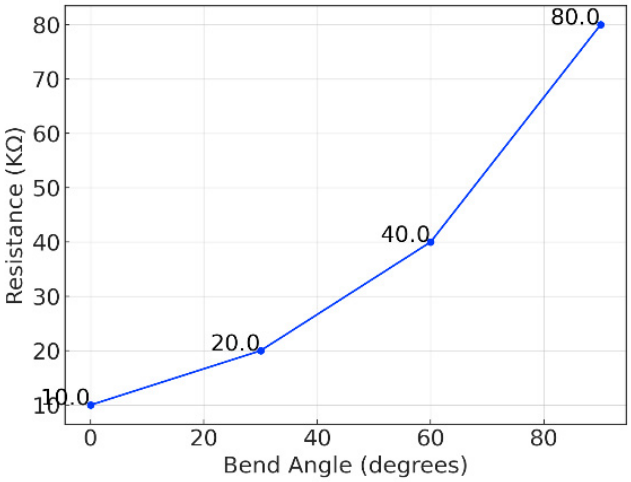

Fig. 4. Angular displacement vs resistance for flex sensors.

Fig. 5 illustrates the relationship between sensor states and recognized gestures, corresponding to specific messages in the sign language recognition system. Each bar represents the state of 4 sensors, detecting different bending positions for gestures A through E. The Y-axis shows the sensor state, where 0 indicates "not bent," 1 indicates "partially bent," and 2 indicates "fully bent." For example, gesture A corresponds to the message "Give Me Water," B represents "Give Me Food," C is "Want to Sleep," D means "I Am Hungry," and E translates to "Need Help." This chart visually explains how different sensor combinations lead to the recognition of distinct hand gestures and their associated messages.

Fig. 6 compares the accuracy of gesture recognition for simple and complex gestures across three users. The blue bars represent the accuracy for simple gestures, ranging from 80% to 90%, while the green bars show the accuracy for complex gestures, ranging from 60% to 80%. The data indicates that the system performs better with simple gestures compared to more complex ones, demonstrating a variation in recognition accuracy depending on gesture complexity.

6. Conclusion

This study successfully demonstrates the conversion of sign language gestures into text, offering an effective communication tool for individuals with speech and hearing impairments. The system achieved high accuracy for simple gestures, while more complex gestures showed slightly lower accuracy, highlighting the need for further refinement in gesture recognition. Testing across multiple users confirmed the system's reliability in real-time

recognition of common sign language gestures. The project contributes to bridging communication gaps, offering a low-cost, practical solution that can be applied in various environments such as education and daily life.

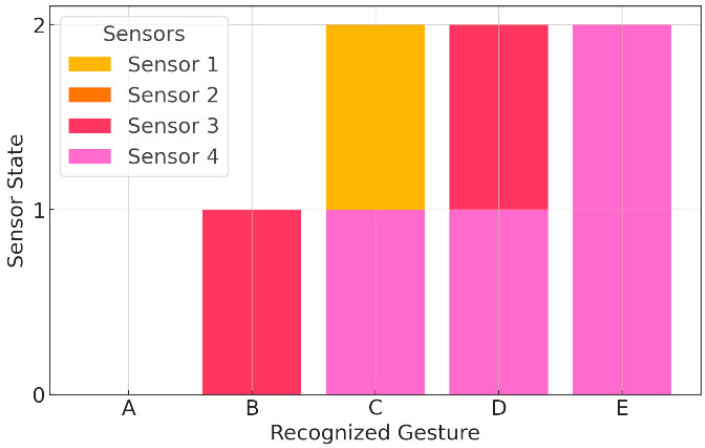

Fig. 5. Sensor state vs. recognized gesture.

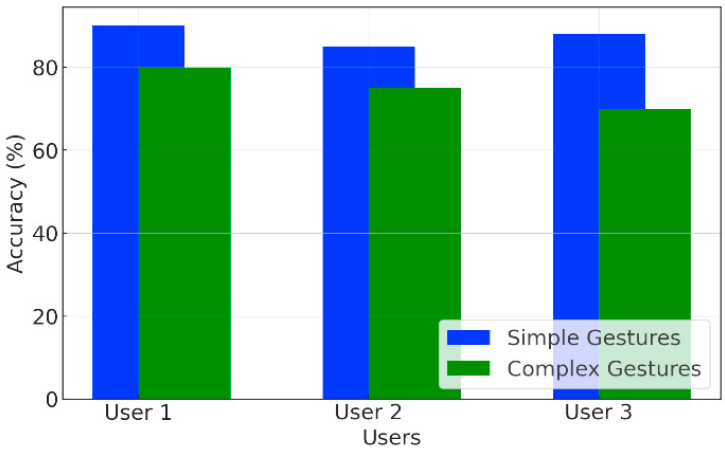

Fig. 6. Comparison of recognition accuracy for simple and complex gestures across users.

Future work could focus on improving recognition accuracy for complex gestures and integrating additional features like voice output, expanding its applicability. Overall, the system presents a step forward in enhancing accessibility for the non-verbal community, fostering greater inclusivity and improving the quality of life for users in social and professional settings.

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